

Watcher Lite 1.1 – tool updates

Feedback from peers included the following items:

1. Change the program to be able to capture port scans from anywhere on the network, not to just the machine running Watcher lite.
2. Cleaner install instructions.
3. Add a requirements.txt file to define requirements for running Watcher lite.
4. Include a readme with quickstart section.
5. Excluding the RFC 5737 (addresses used as documentation and examples) IP addresses to prevent false positives on the scan.

Changes implemented:

1. The program now detects TCP scans (nmap -sS <IP ADDRESS> from anywhere on the local subnet.
2. The program now detects UDP scans (nmap -sU <IP ADDRESS> to the actual machine running Watcher lite.
3. Created a separate document for installation called installation.txt.
4. Created a separate document for program requirements called requirements.txt
5. I believe the readme with a quick start section is covered by the installation.txt file.
6. I debated whether to include the RFC 5737 addresses in the bad_ips.txt file to be excluded from the scan. My thoughts included the debate whether a malicious actor would get on to a local network and use one of those addresses. If that was the case, Watcher lite would then not detect someone using those addresses. I have included them in the bad_ips.txt file. It will be up to the user to decide if they want them excluded from the scan.

Lessons learned

I learned a tremendous amount about python, the development process, and github. I am a new user to github and have only touched the surface of what it is capable of. I also learned a great deal about my classmates' projects. I learned about AI training by reviewing a tool that does federated training. I also learned about Windows event viewer and how to create a windows API in outlook to look for malicious email addresses.

Future work

If I was to continue updating and improving this tool I would add the following things to it.

1. I would add the ability to scan other subnets and networks, not just the one it was on.

2. I would like it to scan for vulnerabilities by connecting to a repository to download known threats (like the IPs it downloads) and scan computers to identify unpatched vulnerabilities.
3. I would like to put together an installer that bundles all the requirements and installs them with one command.